HARSH KHAMKAR

Data Analyst | AI/ML & Data Science Undergraduate

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PROFESSIONAL SUMMARY

AI/ML and Data Science undergraduate with industry experience in data analytics, SQL, Power BI, Zoho Analytics and AI automation. Experienced in building KPI dashboards, analyzing business datasets, processing large-scale data and developing interpretable machine learning workflows. Interested in applying data-driven methods to business and real-world problems.

EDUCATION

B.Tech – Data Science and Artificial Intelligence & Machine Learning — Expected 2028

Universal AI University — **CGPA: 8.17**

TECHNICAL SKILLS

- **Programming & Querying:** Python, SQL, C++, JavaScript
- **Analytics & BI:** Power BI, Zoho Analytics, Advanced Excel, KPI Reporting, Dashboarding, Data Cleaning, Data Visualization
- **Machine Learning & XAI:** Scikit-learn, Predictive Modeling, Model Evaluation, SHAP, Imbalanced Classification
- **Libraries & Platforms:** Pandas, NumPy, Matplotlib, Jupyter Notebook, Git, GitHub, REST APIs
- **Data Engineering & Automation:** Apache Kafka, Hadoop Ecosystem, AiSensy, WhatsApp Automation

PROFESSIONAL EXPERIENCE

Data Analyst & Business Analyst Intern | Choice India (The Joy of Earning) — 18 May 2026 – 18 July 2026

- Analyze business and financial datasets across client accounts (GIC, GFC and MakeMyTrip) using SQL and Zoho Analytics, delivering approximately 3 reports per week.
- Design and maintain 11 interactive Power BI dashboards and automated reports to monitor business KPIs.
- Support AI-powered WhatsApp automation using AiSensy across B2B, B2C and B2G business workflows.
- Clean and process a 3-month MakeMyTrip dataset containing approximately 128,000 records to improve data quality for downstream analysis.

Technology & Innovation Intern | One Globe Build Con LLP, Nagpur — Apr 2025 – Jul 2025

- Built responsive frontend interfaces using HTML, CSS, JavaScript and React.js.
- Contributed to machine learning workflows involving data preprocessing, model evaluation and optimization using Python and Scikit-learn.
- Created analytical dashboards and visualizations using Pandas, Matplotlib, Excel and Power BI.

SELECTED PROJECTS

Explainable Hyperparameter Optimization for Imbalanced Tabular Classification — GitHub

- Developed an explainable, imbalance-aware classification workflow using the UCI Bank Marketing dataset (45,211 records) to study customer term-deposit subscription prediction.
- Evaluated class-weighting and SMOTENC strategies with Optuna-based hyperparameter optimization and multi-objective F1/PR-AUC analysis.
- Implemented threshold optimization, repeated-seed robustness evaluation and SHAP-based feature-importance and explanation-stability analysis.

Student Performance Prediction & Explainability using SHAP — GitHub

- Merged Mathematics and Portuguese student datasets, performed exploratory analysis and engineered an overall academic-performance target.
- Built a Random Forest regression workflow with preprocessing and leakage prevention by excluding intermediate and final grade variables from predictors.
- Applied SHAP to interpret global feature importance and individual predictions, improving model transparency.

Additional work: Selected projects from a broader portfolio of data analytics, AI/ML and software development work. ***Explore GitHub portfolio***

RESEARCH PUBLICATIONS

- **Conference Paper – Under Review:** “SHAP-Based Explainable Student Performance Prediction Using Random Forest, SVM, and Logistic Regression.”
- **Journal Paper – Under Review:** “Explainable Hyperparameter Optimization for Imbalanced Tabular Classification.”

CERTIFICATIONS

Advanced Python | Advanced Excel | Prompt Engineering – DataCamp | SDG – edX (Columbia University) | Climate Change – edX (University of Edinburgh) | HCL Tech

LEADERSHIP & ACHIEVEMENTS

- President, UAI Sports Club (2025–26) – led university sports-event planning, student teams and operations.
- Head of Operations, SMART Project; Event Head, Rush4Rush Intercollege Event.
- National- and State-Level Basketball Player.